

Problem 1

In the fastest measured tennis serve, the ball left the racquet at 73.14 m/s. A served tennis ball is typically in contact with the racquet for 30.0 ms and starts from rest. Assume constant acceleration. (a) What was the ball's acceleration during this serve? (b) How far did the ball travel during the serve?

Given

- $v_i = 0$ m/s — "Starts from rest" means initial velocity is zero
- $v = 73.14$ m/s — Final speed as it leaves the racquet
- $t = 30.0$ ms = 0.0300 s — Convert ms $\rightarrow$ s by dividing by 1000, since equations need SI units

Required

- (a) $a = ?$
(b) $(x - x_i) = ?$ (distance traveled)

Solution

(a) Finding acceleration

Which equation? We are given v_i , v , and t . We are not given — and don't need — displacement $(x - x_i)$. Hence, Eq. 1 is our match.

$$v = v_i + at$$

Isolate a :

$$\begin{aligned} v &= v_i + at \\ v - v_i &= at \\ (v - v_i) / t &= a \\ a &= (v - v_i) / t \end{aligned}$$

Substitute values:

$$\begin{aligned} a &= (73.14 - 0) / 0.0300 \\ a &= 2.44 \times 10^3 \text{ m/s}^2 \end{aligned}$$

This is roughly 249 times the acceleration due to gravity ($g = 9.8 \text{ m/s}^2$) — that's how violent a tennis serve impact really is!

(b) Finding distance traveled

Which equation? We now know v_i , v , t , and a from part (a). We could use Eq. 2, but since a was a rounded answer, reusing it could introduce rounding error. It is safer to use the equation that doesn't need acceleration at all — Eq. 4, which is missing a .

$$(x - x_i) = \frac{1}{2}(v_i + v)t$$

Derivation by transposition — isolate $(x - x_i)$, the distance traveled:

$$\begin{aligned} (x - x_i) &= \frac{1}{2}(v_i + v)t \\ (x - x_i) &= \frac{1}{2}(v_i + v)t \end{aligned}$$

transpose x_i : added on the right, becomes subtracted on the left

Substitute values:

$$\begin{aligned}(x - x_{\text{initial}}) &= \frac{1}{2}(0 + 73.14)(0.0300) \\ (x - x_{\text{initial}}) &= 1.10 \text{ m}\end{aligned}$$

Problem 2

The fastest measured pitched baseball left the pitcher's hand at a speed of 45.0 m/s. If the pitcher was in contact with the ball over a distance of 1.50 m and produced constant acceleration, (a) what acceleration did he give the ball, and (b) how much time did it take him to pitch it?

Given

Symbol	Value	Annotation
v_{initial}	0 m/s	Ball starts at rest in the pitcher's hand before the throwing motion begins
v	45.0 m/s	Speed as the ball leaves the hand
$(x - x_{\text{initial}})$	1.50 m	Distance over which the hand accelerates the ball

Required

- (a) $a = ?$
(b) $t = ?$

Solution

(a) Finding acceleration

Which equation? We are given v_{initial} , v , and $(x - x_{\text{initial}})$. We are not given, and do not yet need, time t . Eq. 3 is the one missing t , so that's our match.

$$v^2 = v_{\text{initial}}^2 + 2a(x - x_{\text{initial}})$$

Derivation by transposition — isolate a :

$$v^2 = v_{\text{initial}}^2 + 2a(x - x_{\text{initial}})$$

$$v^2 - v_{\text{initial}}^2 = 2a(x - x_{\text{initial}})$$

transpose v_{initial}^2 : added on the right, becomes subtracted on the left

$$(v^2 - v_{\text{initial}}^2) / 2(x - x_{\text{initial}}) = a$$

transpose $2(x - x_{\text{initial}})$: multiplying a , becomes a divisor

$$a = (v^2 - v_{\text{initial}}^2) / 2(x - x_{\text{initial}})$$

Substitute values:

$$a = (45.0^2 - 0^2) / 2(1.50) \quad m^2/s^2 \div m$$

$$a = 2025 / 3.00$$

$$a = 675 \text{ m/s}^2$$

(b) Finding time

Which equation? We now know v_i , v , $(x - x_i)$, and a (rounded). To avoid propagating rounding error from part (a), use the equation that doesn't need a — Eq. 4, missing acceleration.

$$(x - x_i) = \frac{1}{2}(v_i + v)t$$

Derivation by transposition — isolate t :

$$(x - x_i) = \frac{1}{2}(v_i + v)t$$

transpose x_i (as in Problem 1)

$$2(x - x_i) = (v_i + v)t$$

transpose $\frac{1}{2}$: multiply both sides by 2 to clear the fraction

$$2(x - x_i) / (v_i + v) = t$$

transpose $(v_i + v)$: multiplying t , becomes a divisor

$$t = 2(x - x_i) / (v_i + v)$$

Substitute values:

$$t = 2(1.50) / (0 + 45.0)$$

$m \div (m/s)$

$$t = 3.00 / 45.0$$

$$t = 0.0667 \text{ s (66.7 ms)}$$

Problem 3

The human body can survive an acceleration trauma incident if the magnitude of the acceleration is less than 250 m/s^2 . If you are in an automobile accident with an initial speed of 105 km/h and are stopped by an airbag, over what minimum distance must the airbag stop you to survive?

Given

Symbol	Value	Annotation
v_i	105 km/h	Must convert to m/s (SI unit) before using the equations
v	0 m/s	The car (and you) come to a complete stop
a	-250 m/s^2	Use the maximum allowed magnitude of deceleration — gives the minimum stopping distance. Negative sign shows deceleration.

Unit conversion:

$$v_i = 105 \text{ km/h} \times (1000 \text{ m} / 1 \text{ km}) \times (1 \text{ h} / 3600 \text{ s}) = 29.17 \text{ m/s}$$

Required

$$(x - x_i) = ? \text{ (minimum stopping distance)}$$

Solution

Which equation? We are given v_i , v , and a . We are not given, and don't need, time t . Eq. 3 is missing t , so that's our match.

$$v^2 = v_i^2 + 2a(x - x_i)$$

Derivation by transposition — isolate $(x - x_i)$:

$$v^2 = v_i^2 + 2a(x - x_i)$$

$$v^2 - v_i^2 = 2a(x - x_i)$$

transpose v_i^2 : added on the right, becomes subtracted on the left

$$(v^2 - v_i^2) / 2a = (x - x_i)$$

transpose $2a$: multiplying, becomes a divisor

$$(x - x_i) = (v^2 - v_i^2) / 2a$$

Substitute values:

$$(x - x_i) = (0^2 - 29.17^2) / 2(-250) \quad m^2/s^2 \div m/s^2$$

$$(x - x_i) = (-850.9) / (-500)$$

$$(x - x_i) = 1.70 \text{ m}$$

The negative signs on top and bottom cancel — that makes sense, since a distance must be positive! The airbag (plus crumple zone) must stop you within at least 1.70 m , or the deceleration will exceed the survivable limit of 250 m/s^2 .

Problem 4

A small block has constant acceleration as it slides down a frictionless incline, released from rest. Its speed after traveling 6.80 m is 3.80 m/s. What is its speed after traveling only 3.40 m (halfway down)?

Strategy. This problem needs two stages. First, use the full trip (0 to 6.80 m) to find the constant acceleration a . Then apply that same a to the shorter trip (0 to 3.40 m) to find the speed at that point.

Given

Symbol	Value	Annotation
v_i	0 m/s	Released from rest at the top
$(x - x_i)_f$	6.80 m	Full distance to the bottom
v_f	3.80 m/s	Speed at the bottom (end of the full distance)
$(x - x_i)_i$	3.40 m	The shorter distance we care about (exactly half of 6.80 m)

Required

$v_i = ?$ — the speed when the block has traveled 3.40 m

Solution

Stage 1 — find the acceleration using the full 6.80 m trip

Which equation? We know v_i , v_f , and $(x - x_i)_f$. We don't have or need t . Eq. 3 is missing t .

$$v_f^2 = v_i^2 + 2a(x - x_i)_f$$

Derivation by transposition — isolate a :

$$v_f^2 = v_i^2 + 2a(x - x_i)_f$$

$$v_f^2 - v_i^2 = 2a(x - x_i)_f$$

transpose v_i^2

$$(v_f^2 - v_i^2) / 2(x - x_i)_f = a$$

transpose $2(x - x_i)_f$: multiplying, becomes a divisor

$$a = (v_f^2 - v_i^2) / 2(x - x_i)_f$$

Substitute values:

$$a = (3.80^2 - 0^2) / 2(6.80) \quad m^2/s^2 \div m$$

$$a = 14.44 / 13.60$$

$$a = 1.06 \text{ m/s}^2 \text{ (unrounded: } 1.0618 \text{ m/s}^2\text{)}$$

Stage 2 — use this acceleration to find the speed at 3.40 m

Same equation, same reasoning. We still don't know or need t , so Eq. 3 applies again — but now we solve for v_i instead of a .

$$v_f^2 = v_i^2 + 2a(x - x_i)_i$$

This is already solved for v_i^2 — one more transposition step (the square root) isolates v_i :

$$v_i = \sqrt{v_f^2 + 2a(x - x_i)_i} \quad \text{apply } \sqrt{\text{ to both sides}}$$

Substitute values (use the un-rounded $a = 1.0618 \text{ m/s}^2$ to avoid rounding error):

$$v_{\blacksquare} = \sqrt{0^2 + 2(1.0618)(3.40)}$$

$$v_{\blacksquare} = \sqrt{7.220}$$

$$v_{\blacksquare} = 2.69 \text{ m/s}$$

Since $v_{\blacksquare} = 0$, speed is proportional to the square root of distance traveled. Half the distance does not give half the speed — it gives speed divided by $\sqrt{2} \approx 1.414$. Check: $3.80 / 1.414 = 2.69 \text{ m/s}$ ✓. Kinematics with acceleration is not "linear" in the intuitive sense!

Problem 5

A Lamborghini Aventador S can go from 0 to 60 mph in 2.7 s. Assume constant acceleration. (a) What is the magnitude of the acceleration? (b) How far has the car traveled when it reaches 60 mph?

Given

Symbol	Value	Annotation
$v_{\blacksquare}$	0 mph = 0 m/s	Starts from rest
v	60 mph	Must convert to m/s
t	2.7 s	Time to reach 60 mph

Unit conversion:

$$v = 60 \text{ mph} \times (1609 \text{ m} / 1 \text{ mi}) \times (1 \text{ h} / 3600 \text{ s}) = 26.8 \text{ m/s}$$

Required

(a) $a = ?$

(b) $(x - x_{\blacksquare}) = ?$

Solution

(a) Finding acceleration

Which equation? We are given $v_{\blacksquare}$, v , and t . We are not given, and don't need, displacement. Eq. 1 is missing $(x - x_{\blacksquare})$, so that's our match.

$$v = v_{\blacksquare} + at$$

Derivation by transposition — isolate a (same steps as Problem 1):

$$v = v_{\blacksquare} + at$$

$$v - v_{\blacksquare} = at \quad \text{transpose } v_{\blacksquare}$$

$$(v - v_{\blacksquare}) / t = a \quad \text{transpose } t$$

$$a = (v - v_{\blacksquare}) / t$$

Substitute values:

$$a = (26.8 - 0) / 2.7 \quad \text{m/s} \div \text{s}$$

$$a = 9.93 \text{ m/s}^2$$

That's about 1.01 g — you'd feel about your own body weight pushing you back into the seat!

(b) Finding distance traveled

Which equation? We know v_i , v_f , t , and a (rounded). To avoid rounding error, use the equation missing a — Eq. 4.

$$(x - x_i) = \frac{1}{2}(v_i + v_f)t$$

Derivation by transposition — isolate $(x - x_i)$ (same steps as Problem 1):

$$(x - x_i) = \frac{1}{2}(v_i + v_f)t$$

Substitute values:

$$(x - x_i) = \frac{1}{2}(0 + 26.8)(2.7)$$

$$(x - x_i) = 36.2 \text{ m}$$

For comparison, that's roughly the length of two and a half basketball courts, covered in under 3 seconds!

Summary of Final Answers

Problem	Part (a)	Part (b)
1. Tennis serve	$a = 2.44 \times 10^3 \text{ m/s}^2$	$(x - x_i) = 1.10 \text{ m}$
2. Baseball pitch	$a = 675 \text{ m/s}^2$	$t = 0.0667 \text{ s}$
3. Airbag	$(x - x_i) = 1.70 \text{ m}$ (minimum)	—
4. Incline block	$v_f = 2.69 \text{ m/s}$ at 3.40 m	—
5. Lamborghini	$a = 9.93 \text{ m/s}^2$	$(x - x_i) = 36.2 \text{ m}$